

Pratik Kubal, M.S.

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Philadelphia, PA | Remote, Hybrid, or Onsite

Senior Software Engineer | Backend & Cloud (AWS) | Full-Stack, DevOps & AI/RAG

Summary

Backend and cloud software engineer with 5+ years building scalable, fault-tolerant services on **AWS**, owning systems from design through production. Led a graph-to-relational **database migration** that cut API latency 90%, re-architected a **Java** processing engine for 2× throughput, and built CI/CD pipelines that shrank deployments from a day to hours. Works full-stack across React micro-frontends and Node.js APIs, and ships AI features: a self-built Claude-API RAG assistant is live at pratik-kubal.com. Revived org-wide code review and mentored engineers.

Technical Skills

Languages & Backend

- **Java, Python**, JavaScript/TypeScript, SQL, Bash – primary backend stack across 5 years in production
- **Node.js**, REST APIs, GraphQL – API design, schema optimization, and integration behind API Gateway
- Microservices & event-driven architecture – Step Functions, SQS; decoupling, fault tolerance, idempotency
- OOP/OOD, dependency injection, TDD – maintainable service design and secure-coding best practices

Cloud & Infrastructure (AWS)

- **AWS Lambda**, Step Functions, Aurora Serverless, DynamoDB, Neptune – serverless compute & data layer
- API Gateway, WAF, SQS/SNS, CloudWatch, EC2, S3, IAM/VPC – networking, security patterns, observability
- **CloudFormation** (IaC) – repeatable, auditable provisioning and automated Graph DB deployments

DevOps & CI/CD

- **GitHub Actions**, AWS CodeDeploy/CodePipeline, Vercel – CI/CD, preview-to-production, security checks
- Firebase, Google Cloud Platform (volunteer) – production deployments and database security hardening

AI Engineering

- **Claude API** (Anthropic), RAG pipelines, vector retrieval – real-time conversational queries over data
- GitHub Copilot, Google Gemini – AI-assisted development and competitive/funding research workflows

Frontend, Testing & Practices

- **React.js**, Next.js, Tailwind CSS, HTML/CSS – responsive UIs, micro-frontends, and SSR API routes
- JUnit, Pytest, Jest, Cypress, Selenium – TDD, integration testing; Agile/Scrum, code review, mentoring

Experience

Cloud Migration, Platform Tooling & Engineering Quality

Sep 2022 – May 2025

Software Engineer II, Dark Matter Technologies

- Led an AWS Neptune to Aurora Serverless migration for a loan-processing platform handling ~100K daily requests, cutting API latency 90% (~1s to sub-100ms), cutting DB load 60%, and saving ~\$10K/year.
- Built a self-service React micro-frontend from scratch in two quarters, enabling two teams to own their configurations and ship independent deployments while cutting configuration deployment time 90%.
- Engineered CI/CD pipelines with GitHub Actions and AWS CodeDeploy that cut deployment cycles from 24 hours to 4 hours, enabling on-demand releases and embedding automated security checks across 3 services.
- Designed event-driven pipelines with AWS Step Functions and SQS to decouple loan-processing microservices, lifting throughput to 100 transactions/min and improving fault tolerance under production load.
- Assisted in rolling out API Gateway and WAF integration patterns that centralized security and standardized API contracts, enabling 6 services to onboard to shared WAF policies and accelerate delivery.
- Engineered Python remediation and alerting scripts that near-eliminated recurring SLA breaches within two weeks, cutting mean time to resolution from ~40 to ~5 minutes and stabilizing client uptime.
- Revived a lapsed monthly cross-team code review program after the senior mentor departed, driving secure-coding and OOP/OOD practices that measurably reduced post-release defects within six months.

- Partnered with Product and Data teams to turn requirements into durable REST APIs and microservices across 6 Agile sprints, shipping 3 features that grew adoption 40% and improved downstream data quality.

Backend Re-Architecture, Performance & Test Engineering

Apr 2020 – Sep 2022

Software Engineer I, Dark Matter Technologies

- Re-architected the Java loan-processing engine with parallel streams and multithreading (mutex/semaphore concurrency controls), achieving 50% faster execution and 2× throughput in production.
- Optimized GraphQL queries and schema design to cut response time ~900ms on key endpoints, and tuned PostgreSQL queries from 1.2s to 300ms (75% faster), reducing latency across core REST services.
- Raised automated test coverage to 70% org-wide by improving Java service testability through dependency injection and building JUnit and Cypress suites that cut regressions and stabilized releases.
- Implemented infrastructure-as-code with AWS CloudFormation for repeatable, auditable environments, and automated Graph DB deployments through Step Functions with 100% deployment reliability.
- Mentored 3 junior developers as their formal mentor, accelerating onboarding and shortening ramp-up by a full sprint, which freed senior capacity and enabled two more project completions across the team.
- Owned on-call rotation and authored runbooks and incident playbooks, instituting postmortems and targeted alerts that prevented nearly 1 in 3 recurring incidents and improved SLO compliance.

Projects

Interactive AI-Driven Resume Assistant (RAG)

Aug 2025

Personal Project, Sole Developer – pratik-kubal.com

- Built a full-stack React/Next.js assistant with a Claude API RAG pipeline that lets recruiters query my resume in real time; tuned Next.js API routes and caching to cut average response latency by ~300ms.
- Implemented end-to-end CI/CD on Vercel with GitHub Actions, cutting deploy time from 30 to 5 minutes; reached 80% coverage with Jest and Cypress and added analytics surfacing top recruiter queries.

Volunteer Experience

Community DevOps & Pro Bono Technology Consulting

Feb 2024 – Present

West Philly Porchfest | Compass Pro Bono

- West Philly Porchfest: stood up a Vercel CI/CD pipeline (per-PR preview builds, merge-to-deploy) for the festival's production website and ran DevOps, hardening Firebase database security against scraping.
- Compass Pro Bono: as Deputy Project Leader of a 6-person team, advised a non-profit on AI adoption, delivering 12 funding-researched recommendations from stakeholder interviews and SWOT analysis.
- Compass Pro Bono: collaborated with a 4-person team to design a Salesforce integration for a non-profit, eliminating 4 of every 5 manual data-entry steps and documenting the workflow for staff.

Education

M.S., Computer Science
University at Buffalo (SUNY), Buffalo, NY

Aug 2018 – Feb 2020
GPA: **3.6/4.0**

B.E., Computer Engineering
University of Mumbai, Mumbai, India

Aug 2013 – Aug 2017

Selected Publication

- “Deep-Learning Compression and Reconstruction of Gigapixel H&E Histopathology Images (TCGA).” Poster, *USCAP 109th Annual Meeting*, 2020 (Abstract #1560). M.S. research, University at Buffalo.

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